

LAB REPORT

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File Carving

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For the class of Computer Forensic, taught by

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1 Introduction

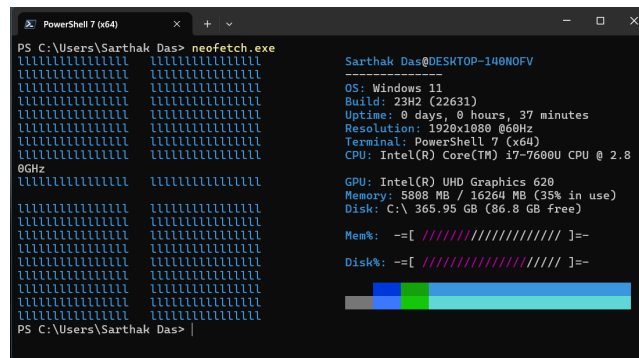
1.1 Objectives

This report is a documentation of my lab work on file carving. The primary objectives of this lab are two-fold:

- File carving on Windows Image.
- Noting the effect of TRIM-enabling.

1.2 System Information

For the purposes of this lab, a Lenovo ThinkPad X270 was used, having the following system specifications: Intel Core i7-7600U CPU @ 2.80GHz with Intel UHD Graphics 620, 16GB RAM, 1TB SSD running operating system Windows 11 23H2 (Build 22631).



```
PS C:\Users\Sarthak Das> neofetch.exe

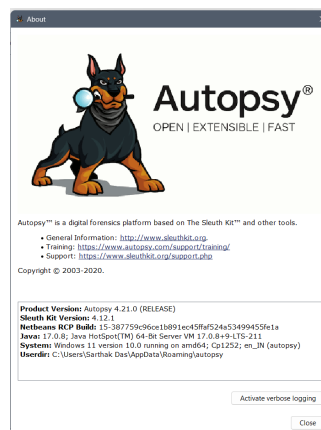
            Sarthak Das@DESKTOP-148NOFV
            -----
            OS: Windows 11
            Build: 23H2 (22631)
            Uptime: 0 days, 0 hours, 37 minutes
            Resolution: 1920x1080 @60Hz
            Terminal: PowerShell 7 (x64)
            CPU: Intel(R) Core(TM) i7-7600U CPU @ 2.8
            GHz
            GPU: Intel(R) UHD Graphics 620
            Memory: 5808 MB / 16264 MB (35% in use)
            Disk: C:\ 365.95 GB (86.8 GB free)

            Mem%:  --[ ##### ]--
            Disk%: --[ ##### ]--

            PS C:\Users\Sarthak Das>
```

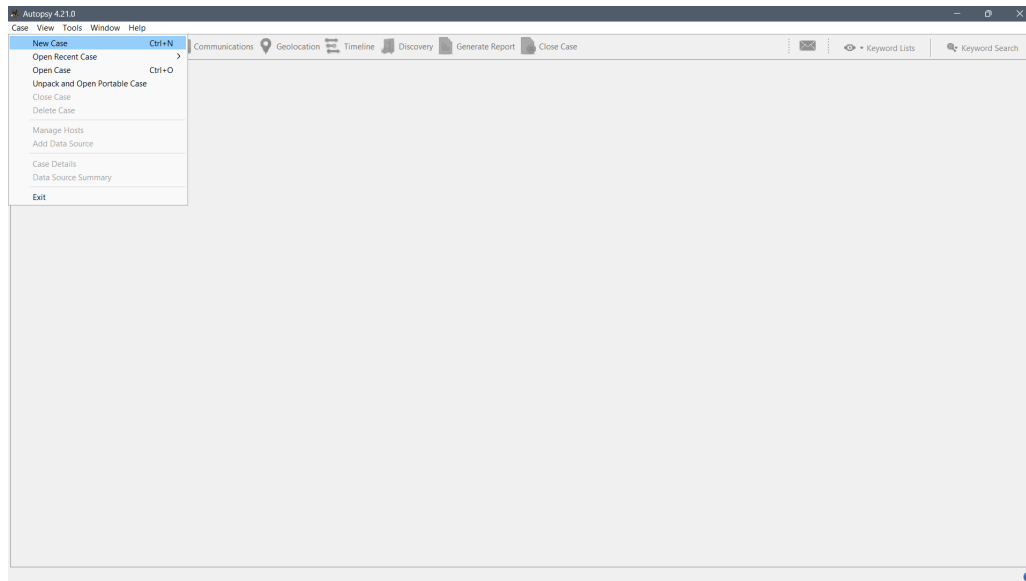
1.3 Software Used

The software used for this practical is Autopsy 4.2.0 (RELEASE).

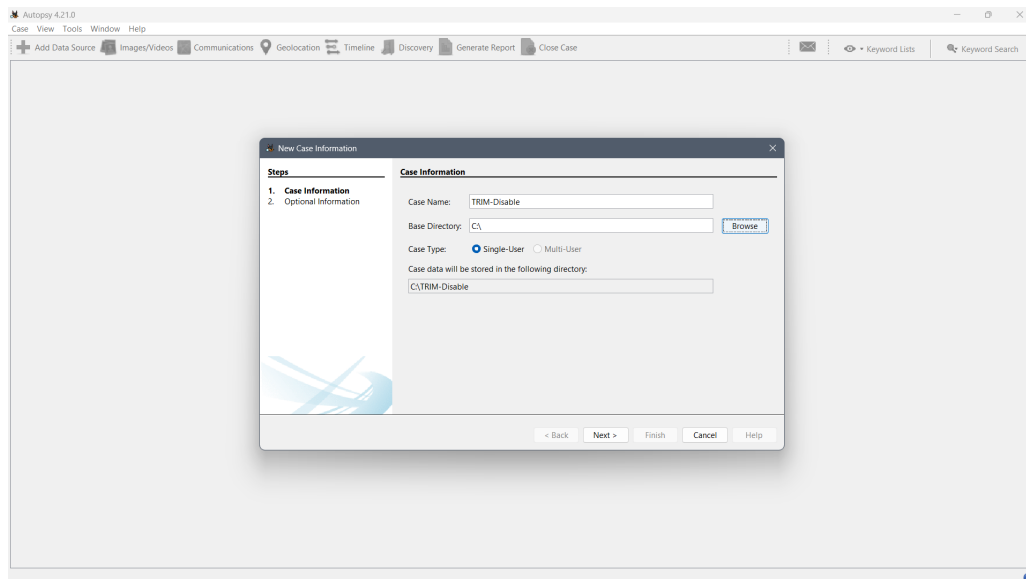


2 Objective 1: File Carving

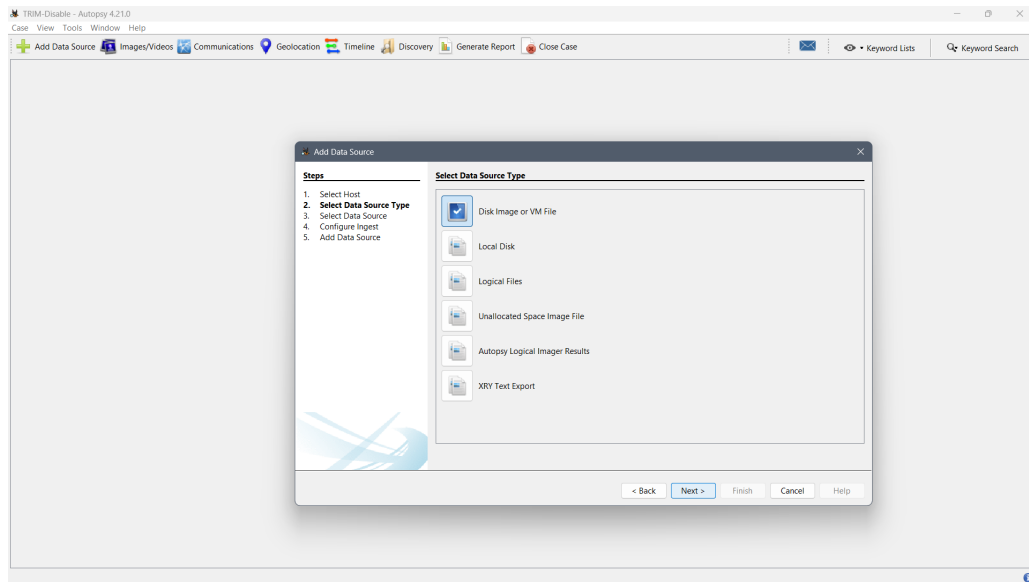
We begin by opening Autopsy, and selecting **Case > New Case**.



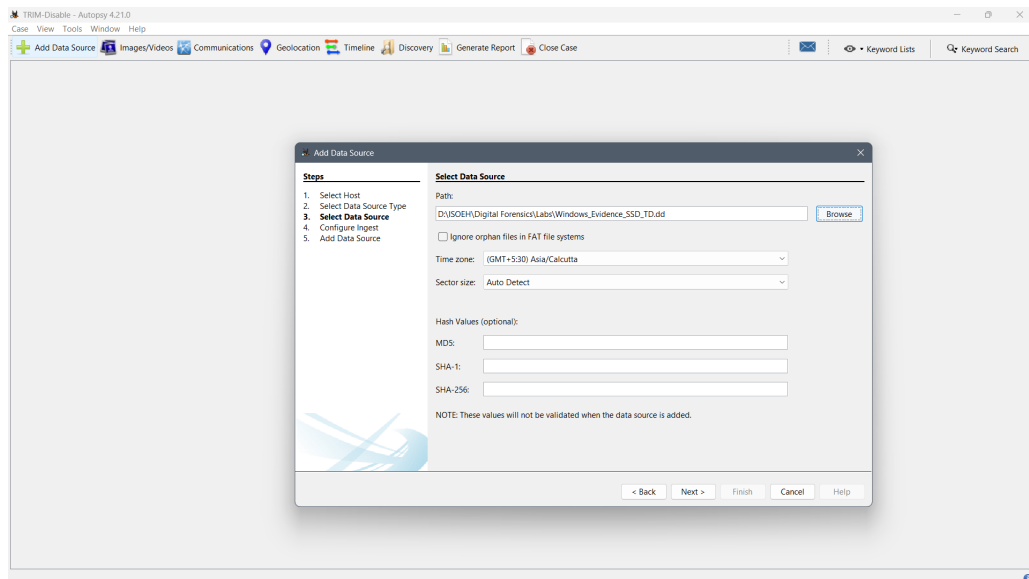
Next, we enter the information such as Case Name, Case Type (single-user or multi-user), and Examiner Information (optional), and click **Finish**.



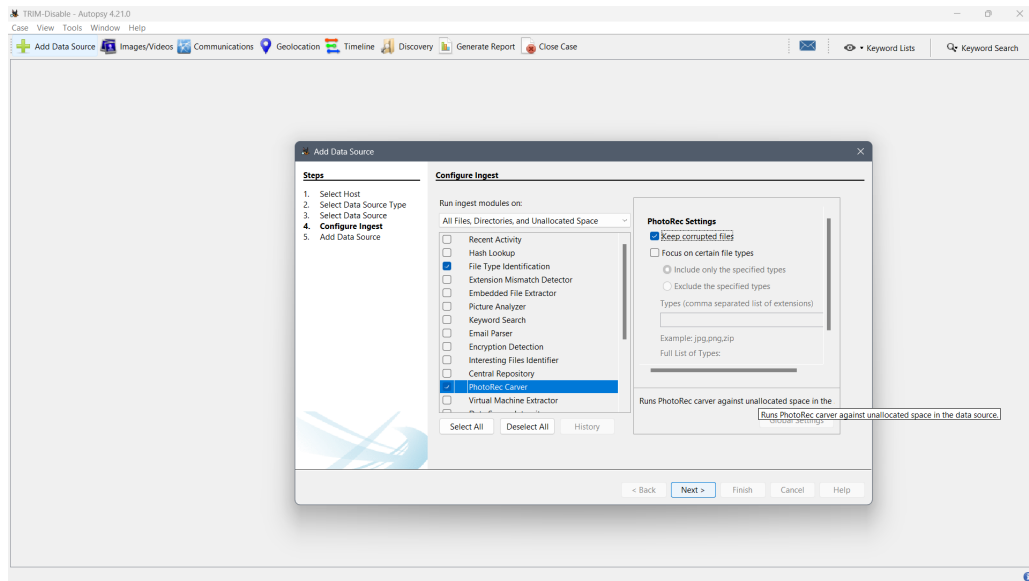
Now we need to add data sources. We select *Add Data Source* and select the data source type as *Disk Image or VM File*.



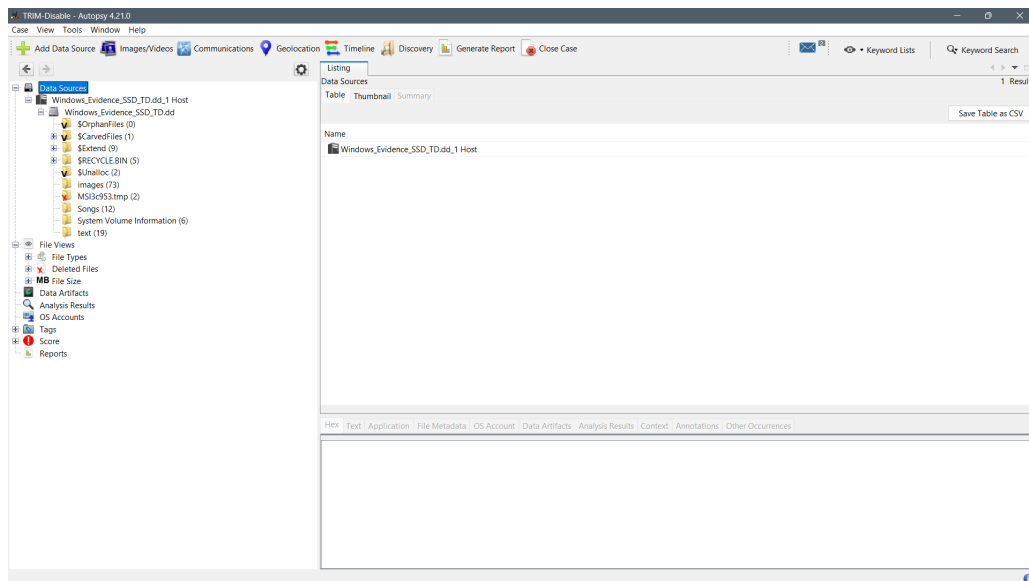
Next, we select the appropriate disk image path. For the purposes of this lab, we were provided with two Windows evidence disk image drives – one TRIM-disabled (1.99GB NTFS) and one TRIM-enabled (1.99GB NTFS).



In the **Configure Ingest** page, we check *File Type Identification* and *PhotoRec Carver* (choosing to keep corrupted files), and finish adding the data source.



Now we are ready to perform analysis. We repeat the process for TRIM-enabled image as well.



3 Objective 2: Effect of TRIM-Enabling

For each case, we click on **Generate Reports** and export it as Microsoft Excel spreadsheets.

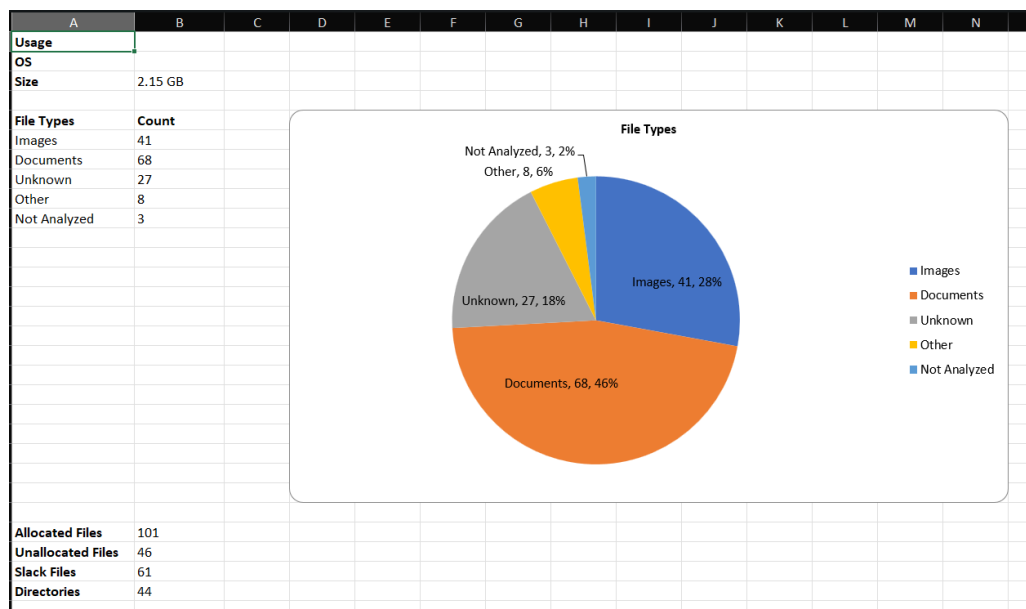


Figure 1: TRIM-disabled image report.

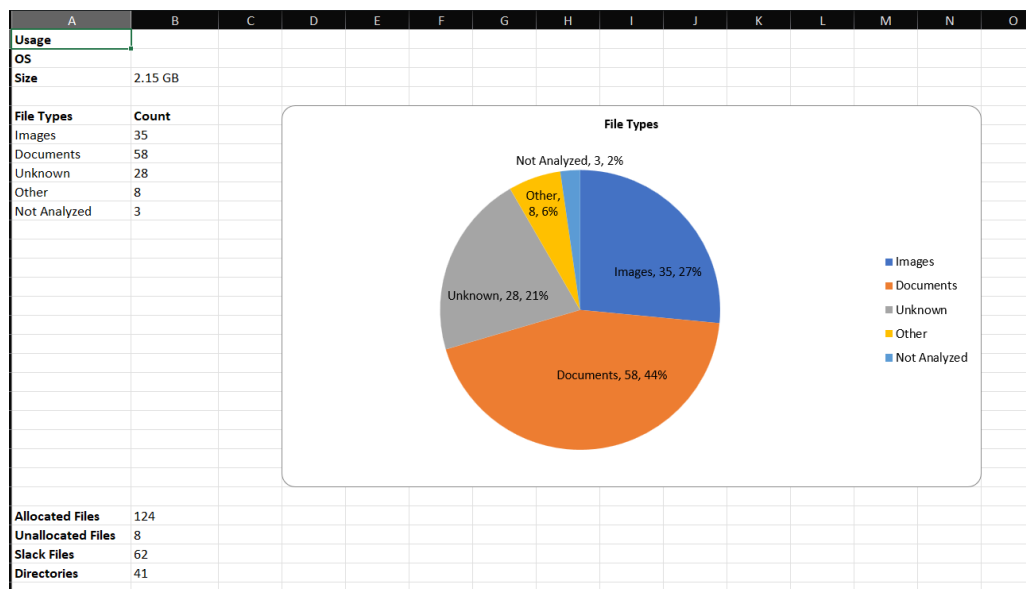


Figure 2: TRIM-enabled image report.

The first thing we notice is that the TRIM-enabled image has only 8 unallocated files as compared to 46 in case of the TRIM-disabled image. Moreover, it only shows 41 directories as compared to 44 in case of the TRIM-disabled image. This is a key result, implying that enabling TRIM causes SSD to zero or otherwise invalidate those blocks on delete, so file carving detects fewer deleted files and directories.

This is also reflected in the following table, which shows that a lesser number of files were recovered from the TRIM-enabled image:

File Type	TRIM-Disabled	TRIM-Enabled
Images	41	35
Documents	68	58
Unknown	27	28
Total	136	121